Factorising Simple Quadratics

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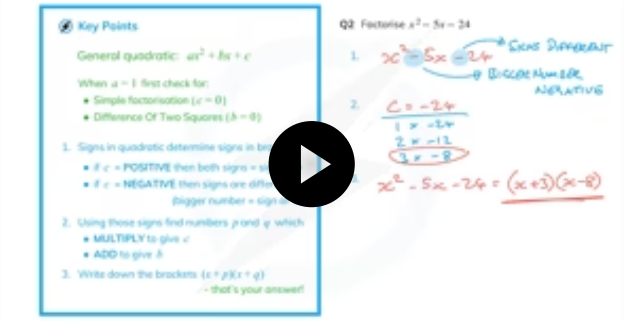

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Factorising simple quadratics

What is a quadratic expression?

- A **quadratic expression** is in the form:
  - $ax^2 + bx + c$ (where $a \neq 0$)
- If there are any **higher powers** of x (like x^3 say) then it is **not** a quadratic

How do I factorise quadratics by inspection?

- This is shown most easily through an example: factorising $x^2 - 2x - 8$
- We need a **pair of numbers** that for $x^2 + bx + c$
  - **multiply** to give c
    - which in this case is -8
  - and **add** to give b
    - which in this case is -2
  - +2 and -4 satisfy these conditions
    - $2 \times (-4) = -8$ and $2 + (-4) = -2$
  - **Write** these numbers in a **pair of brackets** like this:
    - $(x + 2)(x - 4)$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- This is shown most easily through an example: factorising $x^2 - 2x - 8$
- We need a **pair of numbers** that for $x^2 + bx + c$
  - **multiply** to give *c*
    - which in this case is -8
  - and **add** to give *b*
    - which in this case is -2
  - +2 and -4 satisfy these conditions
    - $2 \times (-4) = -8$ and $2 + (-4) = -2$
  - **Rewrite** the middle term by using +2*x* and -4*x*
    - $x^2 + 2x - 4x - 8$
  - **Group** and **factorise** the **first two terms**, using *x* as the common factor
  - and **group** and **factorise** the **last two terms**, using -4 as the common factor
    - $x(x + 2) - 4(x + 2)$
  - Note that these both now have a **common factor** of (*x* + 2) so this **whole bracket** can be factorised out
    - $(x + 2)(x - 4)$

How do I factorise quadratics using a grid?

- This is shown most easily through an example: factorising $x^2 - 2x - 8$
- We need a **pair of numbers** that for $x^2 + bx + c$
  - **multiply** to give *c*
    - which in this case is -8
  - and **add** to give *b*
    - which in this case is -2
  - +2 and -4 satisfy these conditions
    - $2 \times (-4) = -8$ and $2 + (-4) = -2$
  - Write the quadratic equation in a **grid** (as if you had used a grid to expand the brackets)
    - splitting the middle term as +2*x* and -4*x*
- The grid works by multiplying the row and column headings, to give a product in the boxes in the middle

	x^2	-4*x*
	+2*x*	-8

- Write a heading for the first row, using *x* as the highest common factor of x^2 and -4*x*

	$+2x$	-8

- You can then use this to find the headings for the columns
  - e.g. "What does x need to be multiplied by to give x^2?"
  - and "What does x need to be multiplied by to give $-4x$?"

	x	**-4**
x	x^2	$-4x$
	$+2x$	-8

- We can then fill in the remaining row heading using the same idea
  - e.g. "What does x need to be multiplied by to give $+2x$?"
  - or "What does -4 need to be multiplied by to give -8?"

	x	**-4**
x	x^2	$-4x$
$+2$	$+2x$	-8

- We can now read off the factors from the column and row headings
  - $(x+2)(x-4)$

Which method should I use for factorising simple quadratics?

- The first method, by **inspection**, is by far the **quickest**
  - So this is recommended in an exam for simple quadratics (where $a = 1$)
- However some students find the other methods helpful
  - So you may want to learn at least one of them too

Examiner Tips and Tricks

As a check, expand your answer and make sure you get the same expression as the one you were trying to factorise.

Worked Example

(a) Factorise $x^2 - 4x - 21$.

Answer:

give -4
+3 and -7 satisfy this

$$3 \times (-7) = -21$$

$$3 + (-7) = -4$$

Write down the brackets

$$(x + 3)(x - 7)$$

(b) Factorise $x^2 - 5x + 6$.

Answer:

We will factorise by splitting the middle term and grouping

We need two numbers that multiply to 6, and sum to -5
-3 and -2 satisfy this

$$(-3) \times (-2) = 6$$

$$(-3) + (-2) = -5$$

Split the middle term

$$x^2 - 2x - 3x + 6$$

Factorise x out of the first two terms

$$x(x - 2) - 3x + 6$$

Factorise -3 out of the last two terms

$$x(x - 2) - 3(x - 2)$$

These have a common factor of $(x - 2)$ which can be factored out

$$(x - 2)(x - 3)$$

(c) Factorise $x^2 - 2x - 24$.

Answer:

We will factorise by using a grid

We need two numbers that multiply to -24, and sum to -2
+4, and -6 satisfy this

$$4 \times (-6) = -24$$

$$4 + (-6) = -2$$

Use these to split the $-2x$ term and write in a grid

	x^2	$+4x$
	$-6x$	-24

Write a heading using a common factor for the first row

x	x^2	$+4x$

"What does x need to be multiplied by to make x^2?"
"What does x need to be multiplied by to make $+4x$?"

	x	$+4$
x	x^2	$+4x$
	$-6x$	-24

Repeat for the heading for the remaining row
"What does x need to be multiplied by to make $-6x$?"
(Or "What does $+4$ need to be multiplied by to make -24?")

	x	$+4$
x	x^2	$+4x$
-6	$-6x$	-24

Read off the factors from the column and row headings

$$(x + 4)(x - 6)$$

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